OKAN UNIVERSITY / SOFTWARE ENGINEERING

IMAGE PROCESSING PROJECT

INDUSTRIAL SURFACE DEFECT DETECTION

An explainable vision pipeline for automated industrial quality inspection.

- Detect surface defects automatically
- Explain image processing steps
- Compare methods and evaluate results

INDIVIDUAL SEMESTER PROJECT

NOV 2026

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01 THE PROBLEM

Manual product inspection may miss small scratches and cracks, and can lead to inconsistent quality decisions in industrial environments.

02 PROJECT IDEA

Build a system that identifies potential surface defects, highlights their locations, and shows each image-processing step for interpretation. Compare classical image processing methods with an optional AI-based extension.

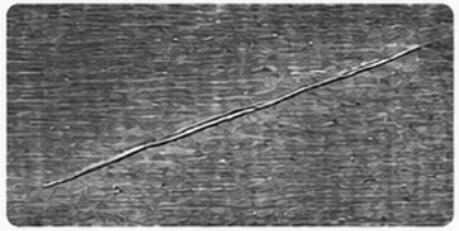
HOW IT WILL WORK

PROPOSED WORKFLOW / CONCEPT DEMO

01 INPUT IMAGE
Upload a product image



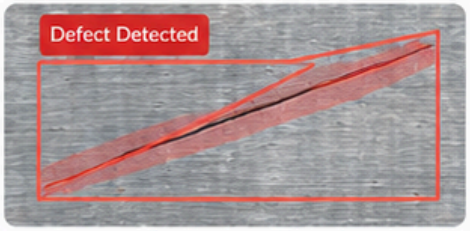
02 PREPROCESSING
Grayscale, denoise, enhance contrast



03 DEFECT DETECTION
Thresholding, edge detection, segmentation



04 RESULT
Highlight detected defects



TECHNICAL SCOPE

01 PREPROCESSING
- Grayscale conversion
- Gaussian Blur (noise reduction)
- Contrast enhancement (CLAHE)

02 SEGMENTATION
- Thresholding
- Canny edge detection
- Morphological operations
- Region analysis

03 AI EXTENSION (OPTIONAL)
- Anomaly detection with deep learning
- Classical vs. AI method comparison
- Compare performance with traditional approaches

04 EVALUATION
- Precision / Recall
- Intersection over Union (IoU)
- Processing time
- Visual results and analysis

DATASET
MVTec AD – selected category, ground-truth masks available.



TECH STACK
Python / OpenCV / NumPy / Streamlit

EXPECTED OUTCOME
A working inspection application that detects and visualizes surface defects, compares image processing methods, and provides a clear analysis of the results.

Dataset reference: mvtec.com/research-teaching/datasets/mvtec-ad | Concept visuals are illustrative; no actual model results are claimed.